

# A certified outer bound for convex NLP scenario subproblems

mpi-sppy

companion to `doc/designs/ipopt_outer_bound_design.md`

## Abstract

The usual device for getting outer bounds in mip-sppy is to read the dual bound off the subproblem solver, which fails for Ipopt, which reports no dual bound. This note derives the bound that the `ipopt_outer_bound` spoke computes instead. The construction is Lagrangian weak duality combined with a tangent-plane underestimator minimised in closed form over the variable box. *None of the mathematics is new*: Theorem 2 is the Frank–Wolfe duality gap on a box, and Section 8 places each result against the literature. The derivation is written out because the hypotheses are what matter in practice, and one of them is easy to get backwards. Its defining property is that it requires *no* assumption about the accuracy of the solve: a truncated or sloppy solve yields a loose bound rather than an invalid one. Convexity, by contrast, is genuinely load-bearing, and Section 5 isolates the one place where it is easy to get backwards.

## 1 Setting

We assume minimization. Fix a scenario  $s$  with probability  $p_s$ ,  $\sum_s p_s = 1$ . Let  $v \in \mathbb{R}^n$  collect *all* variables of the scenario subproblem — the nonanticipative variables  $x(v)$  together with the recourse variables — and let

$$B = \prod_{i=1}^n [l_i, u_i]$$

be the box of variable bounds. With hub weights  $W_s$ , the subproblem an outer bound spoke solves is

$$L_s(W_s) = \min_v \{ f_s(v) + W_s^\top x(v) : g_s(v) \leq 0, h_s(v) = 0, v \in B \}, \quad (1)$$

with  $f_s : \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $g_s : \mathbb{R}^n \rightarrow \mathbb{R}^m$  and  $h_s : \mathbb{R}^n \rightarrow \mathbb{R}^k$ . This is exactly the problem the ordinary Lagrangian spoke builds, with the proximal term switched off.

**Why the returned objective value is useless here.** For a minimisation, the value the solver reports is the objective evaluated at the point it stopped at. Any such value is  $\geq$  the minimum, so it is an *inner* bound: the wrong direction. Nor does Ipopt’s convergence tolerance close the gap, since it is a relative KKT-residual tolerance rather than a bound on the objective error, and the two error sources point opposite ways — optimality error stops slightly above the optimum, while constraint violation can place the iterate slightly below it.

## 2 Weak duality

**Lemma 1** (Weak duality). *Let  $\lambda \in \mathbb{R}_+^m$  and  $\mu \in \mathbb{R}^k$  be arbitrary. Define*

$$\varphi_s(v) = f_s(v) + W_s^\top x(v) + \lambda^\top g_s(v) + \mu^\top h_s(v), \quad (2)$$

$$q_s(\lambda, \mu) = \inf_{v \in B} \varphi_s(v). \quad (3)$$

*Then  $q_s(\lambda, \mu) \leq L_s(W_s)$ .*

*Proof.* Let  $v$  be any point feasible for (1). Then  $v \in B$ , so  $q_s(\lambda, \mu) \leq \varphi_s(v)$ . Moreover  $g_s(v) \leq 0$  and  $\lambda \geq 0$  give  $\lambda^\top g_s(v) \leq 0$ , and  $h_s(v) = 0$  gives  $\mu^\top h_s(v) = 0$ ; hence  $\varphi_s(v) \leq f_s(v) + W_s^\top x(v)$ . Chaining the two inequalities,  $q_s(\lambda, \mu) \leq f_s(v) + W_s^\top x(v)$  for every feasible  $v$ . Taking the infimum over feasible  $v$  gives the claim.  $\square$

Note what Lemma 1 does *not* require:  $\lambda$  and  $\mu$  need not be optimal, or even good. Any  $\lambda \geq 0$  and any  $\mu$  will do. This is the property that makes the whole approach robust, and it is why a mistaken sign convention or a stale multiplier can only cost tightness (Remark 4).

**The trap.** Lemma 1 is not yet usable, because  $q_s$  is defined by an *infimum*. Handing  $\varphi_s$  to a nonlinear solver returns a *point*, and the value there is  $\geq$  the infimum — an upper bound on  $q_s$ , which is the wrong direction again. A second NLP solve therefore does not by itself certify anything.

## 3 The box underestimator

**Theorem 2** (Certified bound). *Suppose  $\varphi_s$  is convex on an open set containing  $B$  and differentiable at  $\hat{v}$ . Define*

$$\hat{q}_s(\hat{v}) = \varphi_s(\hat{v}) + \sum_{i=1}^n \min_{t \in [l_i, u_i]} \partial_i \varphi_s(\hat{v}) (t - \hat{v}_i). \quad (4)$$

*Then  $\hat{q}_s(\hat{v}) \leq q_s(\lambda, \mu) \leq L_s(W_s)$ , and each term of the sum is available in closed form:*

$$\min_{t \in [l_i, u_i]} \partial_i \varphi_s(\hat{v}) (t - \hat{v}_i) = \begin{cases} \partial_i \varphi_s(\hat{v}) (l_i - \hat{v}_i), & \partial_i \varphi_s(\hat{v}) > 0, \\ \partial_i \varphi_s(\hat{v}) (u_i - \hat{v}_i), & \partial_i \varphi_s(\hat{v}) < 0, \\ 0, & \partial_i \varphi_s(\hat{v}) = 0. \end{cases} \quad (5)$$

*Proof.* Convexity and differentiability at  $\hat{v}$  give the gradient inequality

$$\varphi_s(v) \geq \varphi_s(\hat{v}) + \nabla \varphi_s(\hat{v})^\top (v - \hat{v}) \quad \text{for all } v.$$

Minimising both sides over  $v \in B$  preserves the inequality, and the right-hand side is separable across coordinates because  $B$  is a box, which yields (4). Each one-dimensional problem minimises a linear function over an interval, so the minimum is attained at an endpoint, giving (5). The final inequality is Lemma 1.  $\square$

**Corollary 3.**  $\hat{q}_s(\hat{v}) \leq L_s(W_s)$  for any  $\hat{v}$  at which  $\varphi_s$  is differentiable, any  $\lambda \geq 0$  and any  $\mu$ . In particular  $\hat{v}$  need not be optimal, and need not even be feasible for (1).

Corollary 3 is the whole point. The certificate consumes the iterate the solver happened to stop at and the multipliers it happened to report, and returns a valid bound regardless. Computationally it costs one gradient evaluation and a loop over the variables — no second solve.

*Remark 4 (Robustness).* Because Corollary 3 holds for arbitrary  $\lambda \geq 0$  and  $\mu$ , an error in the sign convention used to recover the multipliers, a stale multiplier, or a multiplier clipped to zero can only make  $\hat{q}_s$  smaller. Such an error makes the bound loose, never invalid. Convexity enjoys no such protection; see Section 5.

## 4 Exactness at a KKT point

The correction term in (4) is not merely small in practice: it vanishes identically at an exact KKT point, so the certificate degrades gracefully and costs nothing when the solve is good.

**Proposition 5** (The correction vanishes). *Let  $\hat{v}$  satisfy the KKT conditions of (1) with multipliers  $\lambda \geq 0$  for  $g_s \leq 0$ ,  $\mu$  for  $h_s = 0$ , and  $z_L, z_U \geq 0$  for the bounds  $l - v \leq 0$  and  $v - u \leq 0$ . Then every term of the sum in (4) is zero, and if in addition  $\varphi_s$  is convex then  $\hat{q}_s(\hat{v}) = L_s(W_s)$ .*

*Proof.* Stationarity of the full Lagrangian, which is  $\varphi_s$  augmented with the bound terms, reads

$$\nabla \varphi_s(\hat{v}) = z_L - z_U. \quad (6)$$

Fix a coordinate  $i$  and consider the three possible positions of  $\hat{v}_i$ .

If  $l_i < \hat{v}_i < u_i$ , complementarity gives  $z_{L,i} = z_{U,i} = 0$ , so  $\partial_i \varphi_s(\hat{v}) = 0$  by (6) and the term is 0 by (5).

If  $\hat{v}_i = l_i$ , complementarity gives  $z_{U,i} = 0$ , so  $\partial_i \varphi_s(\hat{v}) = z_{L,i} \geq 0$ . When the derivative is strictly positive (5) selects  $t = l_i = \hat{v}_i$ , and when it is zero the term is zero outright; either way the term is 0.

If  $\hat{v}_i = u_i$ , symmetrically  $z_{L,i} = 0$  and  $\partial_i \varphi_s(\hat{v}) = -z_{U,i} \leq 0$ , so (5) selects  $t = u_i = \hat{v}_i$  and the term is 0.

Hence  $\hat{q}_s(\hat{v}) = \varphi_s(\hat{v})$ . Complementarity  $\lambda^\top g_s(\hat{v}) = 0$  and feasibility  $h_s(\hat{v}) = 0$  reduce (2) to  $\varphi_s(\hat{v}) = f_s(\hat{v}) + W_s^\top x(\hat{v})$ . Under convexity the KKT conditions are sufficient for global optimality of (1), so that value is  $L_s(W_s)$ .  $\square$

Proposition 5 has a practical corollary worth stating plainly: the correction term *measures its own inexactness*. It is zero when the solve is exact and grows as the solve degrades, so the reported bound loosens smoothly rather than becoming wrong.

## 5 Canonical form, and where convexity is easy to get backwards

Theorem 2 requires  $\varphi_s$  convex, and by (2) that requires each component of  $g_s$  to be convex and each component of  $h_s$  to be affine, since  $\lambda \geq 0$  but  $\mu$  is free in sign.

The subtlety is that  $g_s$  is the constraint in *canonical* form  $g_s(v) \leq 0$ , and putting a  $\geq$  row into canonical form negates its body. Writing  $b(v)$  for the constraint body as the user stated it:

| as written                    | canonical form   | requirement on $b$ |
|-------------------------------|------------------|--------------------|
| $b(v) \leq \beta$             | $g = b - \beta$  | $b$ convex         |
| $b(v) \geq \alpha$            | $g = \alpha - b$ | $b$ <b>concave</b> |
| $\alpha \leq b(v) \leq \beta$ | both rows        | $b$ affine         |
| $b(v) = \beta$                | $h = b - \beta$  | $b$ affine         |

So  $x^2 \leq 4$  is admissible and  $x^2 \geq 1$  is not, though both are written with a convex body — and indeed the feasible set of the second is not convex. This is the one place in the construction where an entirely ordinary-looking model silently leaves the hypotheses, and the consequence is not a loose bound but an invalid one. A worked instance: for

$$\min x \quad \text{s.t.} \quad x^2 \geq 1, \quad x \in [0, 1.5],$$

whose optimum is 1, taking  $\hat{v} = 0.5$  and  $\lambda = 1$  in (4) yields  $\hat{q} = 1.25 > 1$ .

Of the four rows above, three are decidable — affineness is a question about polynomial degree — and the implementation enforces them. Convexity or concavity of a one-sided nonlinear body is not decidable, and remains a user assertion.

## 6 Aggregating across scenarios

The per-scenario bounds are combined by the usual probability-weighted sum, and the condition under which that sum is itself a bound deserves care.

**Proposition 6** (Aggregation). *Let OPT be the optimal value of the original stochastic program. If the weights satisfy  $\sum_s p_s W_s = 0$ , then  $\sum_s p_s L_s(W_s) \leq \text{OPT}$ .*

*Proof.* Let  $x^*$  together with the scenario recourse decisions be optimal for the original problem, so that  $\text{OPT} = \sum_s p_s f_s(v_s^*)$  where  $v_s^*$  is the optimal decision in scenario  $s$  and  $x(v_s^*) = x^*$  for every  $s$  by nonanticipativity. Each  $v_s^*$  is feasible for subproblem (1), so  $L_s(W_s) \leq f_s(v_s^*) + W_s^\top x^*$ . Multiplying by  $p_s$  and summing,

$$\sum_s p_s L_s(W_s) \leq \sum_s p_s f_s(v_s^*) + \left( \sum_s p_s W_s \right)^\top x^* = \text{OPT} + 0. \quad \square$$

*Remark 7* (Weights must share a generation). The hypothesis  $\sum_s p_s W_s = 0$  is a *joint* condition on the whole weight vector, and progressive hedging maintains it at each iteration. It is not preserved by mixing: if  $W'_s$  takes scenario  $s$ 's weights from one iteration and scenario  $t$ 's from another, then in general  $\sum_s p_s W'_s \neq 0$ , the proof above leaves the residual term  $(\sum_s p_s W'_s)^\top x^*$  in place, and the resulting number is not a bound at all. Nothing downstream can detect this, which is why a scenario whose certificate is unavailable must make the whole expectation unavailable rather than contribute a stale value.

## 7 Practical consequences

**Tightness is governed by the box.** By (5) the shortfall of  $\hat{q}_s$  below  $\varphi_s(\hat{v})$  is

$$\sum_{i=1}^n |\partial_i \varphi_s(\hat{v})| \cdot (\text{distance from } \hat{v}_i \text{ to the far end of } [l_i, u_i]),$$

so the bound is only as tight as the variable bounds are. Gradient components at a converged solve are small but generically nonzero, so a box of width 10 costs little while a box of width  $10^{10}$  can cost a substantial fraction of the objective. Tightening bounds before certifying — for instance by feasibility-based bounds tightening, which shrinks  $B$  without removing feasible points and therefore only raises  $\hat{q}_s$  — is the single most effective lever on quality.

**Unbounded variables.** If  $l_i = -\infty$  and  $\partial_i \varphi_s(\hat{v}) > 0$ , or  $u_i = +\infty$  and  $\partial_i \varphi_s(\hat{v}) < 0$ , the corresponding term in (5) is  $-\infty$  and no finite bound is available. Whether this occurs depends on the model and on how converged the solve is: at an exact KKT point the offending component is zero by Proposition 5, but away from one it generally is not. The implementation reports “no bound” in that case rather than  $-\infty$ , so that Remark 7 is respected.

**Floating point.** Nothing in Theorem 2 requires a tolerance. A small relative cushion, reporting  $\hat{q}_s - \varepsilon(1 + |\hat{q}_s|)$ , is nevertheless cheap insurance against last-bit effects in the gradient and the sum; it is hygiene, not part of the argument.

## 8 Relation to known results

Nothing in Sections 2–6 is new, and it is worth saying which wheel each result re-uses.

**Lemma 1** is textbook Lagrangian weak duality, stated for an arbitrary multiplier pair rather than an optimal one.

**Theorem 2** is the *Frank–Wolfe duality gap*, sometimes called the Wolfe gap or the linearisation gap. For a convex  $f$  on a compact convex  $D$ , evaluating the linear minimisation oracle at  $x$  gives

$$f(x) + \min_{y \in D} \nabla f(x)^\top (y - x) \leq \min_D f,$$

which is (4) with  $D = B$ . The bound is as old as the Frank–Wolfe algorithm itself [1]; its modern reading as a *certificate* computable at any iterate, rather than as a step-size device, is the framing in [2]. The only specialisation here is that a box makes the oracle separable and closed-form, so the certificate costs one gradient evaluation and no optimisation at all.

**Proposition 6** is the standard argument by which progressive hedging weights yield a lower bound on the optimal value, as in [3].

**The combination** of progressive hedging with Frank–Wolfe machinery to obtain Lagrangian dual bounds is itself established [4], and mpi-sppy already ships an FWPH cylinder on that basis.

What is specific here is therefore an assembly rather than a result: the multipliers are the ones Ipopt already returns, the linearisation point is the iterate it already stopped at, and the box is the model’s own variable bounds — so a valid outer bound is obtained from the solve the spoke was going to perform anyway, with no second optimisation and no assumption that the first one converged. The wider question of extracting valid dual bounds from an inexact oracle is treated at length in the inexact bundle-method literature, which this note does not attempt to survey.

## 9 Correspondence with the implementation

| object here          | in the code                                        |
|----------------------|----------------------------------------------------|
| $\varphi_s$ in (2)   | <code>_lagrangian_expression</code>                |
| $\hat{q}_s$ in (4)   | <code>certified_lower_bound</code>                 |
| decidable hypotheses | <code>check_model_is_certifiable</code>            |
| box tightening       | <code>unbounded_variables</code>                   |
| Proposition 6 sum    | <code>SP0pt.Ebound</code>                          |
| Remark 7 safeguard   | a <code>None</code> bound, propagated collectively |

The first four live in `mpisppy/utils/dual_certificate.py`; the spoke that drives them is `mpisppy/cylinders/ipopt_outer_bound.py`.

## References

- [1] Frank, M. and Wolfe, P.: An algorithm for quadratic programming. *Naval Research Logistics Quarterly* **3**(1–2), 95–110 (1956).
- [2] Jaggi, M.: Revisiting Frank–Wolfe: projection-free sparse convex optimization. *Proceedings of the 30th International Conference on Machine Learning (ICML)*, 427–435 (2013).
- [3] Gade, D., Hackebeil, G., Ryan, S.M., Watson, J.-P., Wets, R.J.-B. and Woodruff, D.L.: Obtaining lower bounds from the progressive hedging algorithm for stochastic mixed-integer programs. *Mathematical Programming* **157**(1), 47–67 (2016).
- [4] Boland, N., Christiansen, J., Dandurand, B., Eberhard, A., Linderoth, J., Luedtke, J. and Oliveira, F.: Combining progressive hedging with a Frank–Wolfe method to compute Lagrangian dual bounds in stochastic mixed-integer programming. *SIAM Journal on Optimization* **28**(2), 1312–1336 (2018).